# 深圳市凯亿星电子有限公司

| | |
|---|---|
| Project No.<br>项目编号 | S1918-A101KL68-42W.D |
| Customer<br>客户名称 | |
| Module No.<br>客户型号 | |
| Product type<br>产品内容 | Standard LCM Switch FPC<br>800RGB x 1280 Dots<br>10.1”TFT LCD |
| Signature by customer:<br>客户确认签章： | |

| 编　　制 | 电子审核 | 结构审核 | 批　　准 |
|---|---|---|---|
| | | | |

# REVISION STATUS

| Version | Revise Date | Page | Content | Modified by |
|---|---|---|---|---|
| V1.0 | 2018.03.08 | - | First Issued. | DSR |

# TABLE OF CONTENTS


| NO. | CONTENTS | PAGE |
|---|---|---|

REVISION STATUS ..... 2
TABLE OF CONTENTS ..... 3
1. GENERAL DESCRIPTION ..... 4
2. MECHANICAL SPECIFICATION ..... 5
3. PIN DESCRIPTION ..... 6
4. ELECTRICAL CHARACTERISTICS ..... 7
5. SEQUENTIAL CHART ..... 10
6.OPTICAL CHARACTERISTICS ..... 12
7.RELIABILITY TEST ITEMS ..... 15
8. GENERAL PRECAUTION ..... 16
9. PACKAGE DRAWING ..... 16

# 1. GENERAL DESCRIPTION

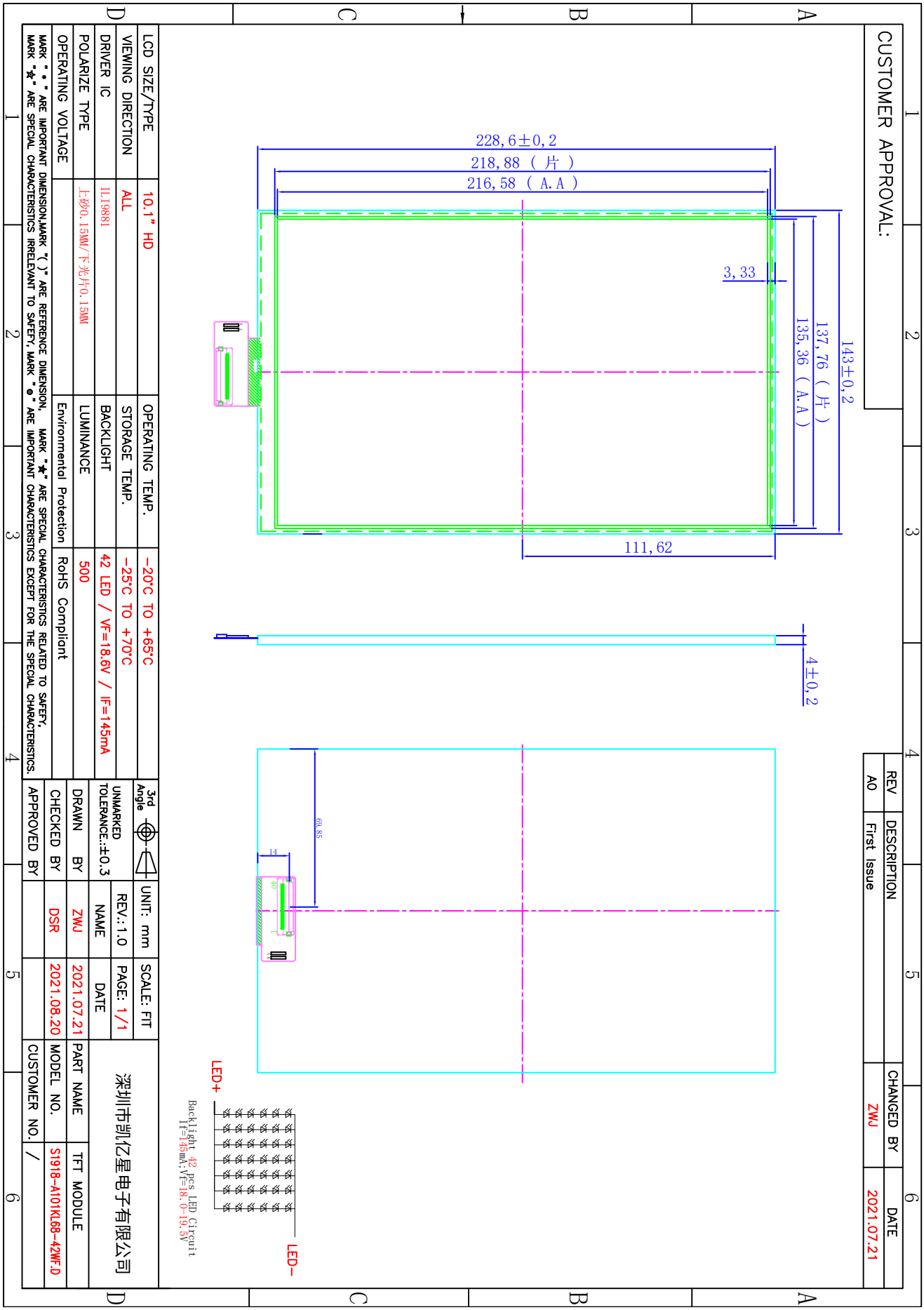
## 1.1 DESCRIPTION

S1918-A101KL68-42W.D is a color active matrix thin film transistor (TFT) IPS liquid crystal display (LCD) that uses amorphous silicon TFT as a switching device. It is composed of a TFT LCD panel, Driver IC ,FPC and Backlight.

## 1.2 FEATURES:

| No. | Item | Specification | Unit |
|---|---|---|---|
| 1 | Panel Size | 10.1” | inch |
| 2 | Number of Pixels | 800×RGB (3) ×1280 | pixels |
| 3 | Active Area | 135.36(H)x 216.58 (V) | mm |
| 4 | Pixel Pitch | 0.0564 (H)×0.1692(V) | mm |
| 5 | Outline Dimension | 143.06(W)×228.66(H)×4.00(D) | mm |
| 6 | Number of Colors | 16.7M | - |
| 7 | Display Mode | Normally Black | - |
| 8 | Viewing Direction | ALL | - |
| 9 | Display Format | RGB vertical stripe | - |
| 10 | Luminance (cd/m^2) | 500(TYP.) | nit |
| 11 | Contrast Ratio | 800(TYP.) | |
| 12 | Surface Treatment | Anti-Glare | - |
| 13 | Interface | MIPI | - |
| 14 | Backlight | White LED | - |
| 15 | Operation Temperature | -20~70 | ℃ |
| 16 | Storage Temperature | -25~75 | ℃ |

# 2. MECHANICAL SPECIFICATION




| | | | |
|---|---|---|---|
| LCD SIZE/TYPE | 10.1" HD | OPERATING TEMP. | -20°C TO +65°C |
| VIEWING DIRECTION | ALL | STORAGE TEMP. | -25°C TO +70°C |
| DRIVER IC | IL19881 | BACKLIGHT | 42 LED / VF=18.6V / IF=145mA |
| POLARIZE TYPE | 上砂0.15MM/下光片0.15MM | LUMINANCE | 500 |
| OPERATING VOLTAGE | | Environmental Protection | RoHS Compliant |

MARK "( )" ARE REFERENCE DIMENSION. MARK "★" ARE SPECIAL CHARACTERISTICS RELATED TO SAFEFY.
MARK "★" ARE SPECIAL CHARACTERISTICS IRRELEVANT TO SAFEFY, MARK "●" ARE IMPORTANT CHARACTERISTICS EXCEPT FOR THE SPECIAL CHARACTERISTICS.

| | | | | | |
|---|---|---|---|---|---|
| 3rd Angle | UNIT: mm | SCALE: FIT | 深圳市凯亿星电子有限公司 | | |
| UNMARKED TOLERANCE:±0.3 | REV.:1.0 | PAGE: 1/1 | | | |
| DRAWN BY | NAME | DATE | PART NAME | TFT MODULE | |
| ZWJ | 2021.07.21 | | | | |
| CHECKED BY | DSR | 2021.08.20 | MODEL NO. | S1918-A101KL68-42WF.D | |
| APPROVED BY | | | CUSTOMER NO. | / | |

# 3. PIN DESCRIPTION

| Pin No. | Symbol | Type | Function |
|---|---|---|---|
| 1 | NC | - | No connection |
| 2 | VDD | P | Power supply 3.3V |
| 3 | VDD | P | Power supply 3.3V |
| 4 | GND | P | Ground |
| 5 | Reset | I | Global reset pin(3.3V) |
| 6 | NC | - | No connection |
| 7 | GND | P | Ground |
| 8 | MIPI-0N | I/O | MIPI Date negative signal |
| 9 | MIPI-0P | I/O | MIPI Date Positive signal |
| 10 | GND | P | Ground |
| 11 | MIPI-1N | I | MIPI Date negative signal |
| 12 | MIPI-1P | I | MIPI Date Positive signal |
| 13 | GND | P | Ground |
| 14 | MIPI-CLKN | I | MIPI CLK negative signal |
| 15 | MIPI-CLKP | I | MIPI CLK Positive signal |
| 16 | GND | P | Ground |
| 17 | MIPI-2N | I | MIPI Date negative signal |
| 18 | MIPI-2P | I | MIPI Date Positive signal |
| 19 | GND | P | Ground |
| 20 | MIPI-3N | I | MIPI Date negative signal |
| 21 | MIPI-3P | I | MIPI Date Positive signal |
| 22 | GND | P | Ground |
| 23 | NC | - | No connection |
| 24 | NC | - | No connection |
| 25 | GND | P | Ground |
| 26 | NC | - | No connection |
| 27 | NC | - | No connection |
| 28 | NC | - | No connection |
| 29 | NC | - | No connection |
| 30 | GND | P | Ground |
| 31 | LED- | P | Power for LED backlight (Cathode) |
| 32 | LED- | P | Power for LED backlight (Cathode) |
| 33 | NC | - | No connection |
| 34 | NC | - | No connection |
| 35 | NC | - | No connection |
| 36 | NC | - | No connection |
| 37 | NC | - | No connection |
| 38 | NC | - | No connection |
| 39 | LED+ | P | Power for LED backlight (Anode) |
| 40 | LED+ | P | Power for LED backlight (Anode) |

# 4. ELECTRICAL CHARACTERISTICS

## 4.1 ABSOLUTE MAXIMUM RATINGS

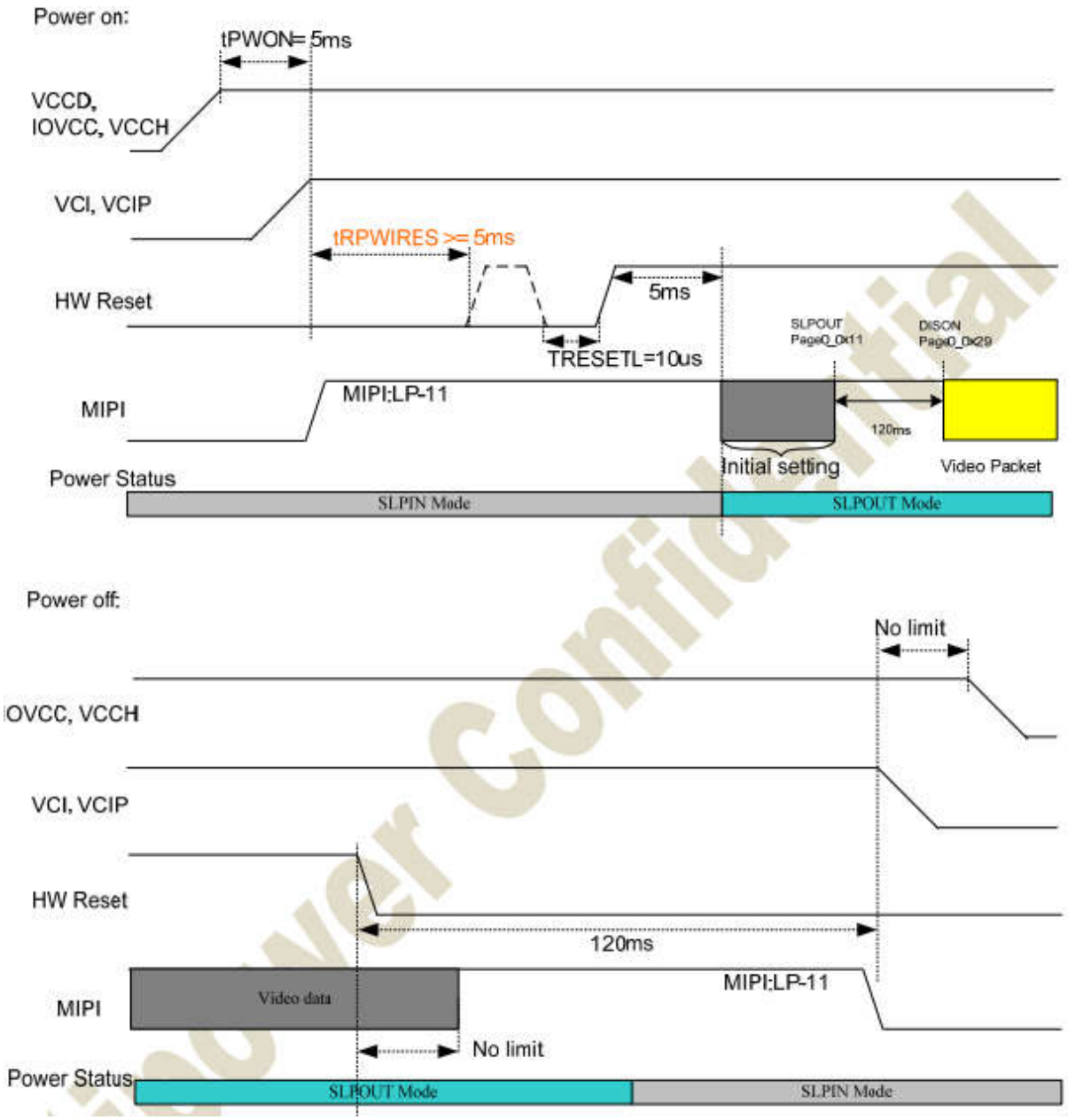
| Item | Symbol | Min. | Max. | Unit | Conditions |
|---|---|---|---|---|---|
| Power Voltage | VDD | -0.3 | 5.0 | V | |
| Input Signal Voltage | $V_I$ | -0.3 | VDD | V | |

## 4.2 TFT LCD MODULE

### 4.2.1 Operating Conditions

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Conditions |
|---|---|---|---|---|---|---|
| Power Voltage | VDD | 3.0 | 3.3 | 3.6 | V | |
| Negative LC voltage | AVEE | - | -5.5 | - | V | |
| Postive LC voltage | AVDD | - | 5.5 | - | V | |
| Current Consumption | $I_{VDD}$ | - | - | 125 | mA | |

## 4.3 Power ON/OFF Sequence




Power Confidential

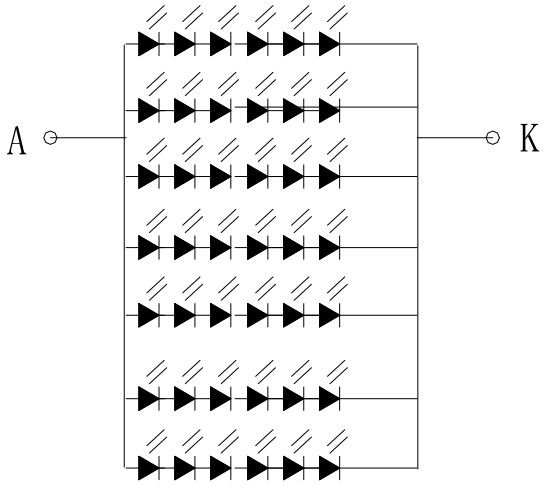
## 4.4 BACK LIGHT UNIT

| PARAMETER | Sym. | Min. | Typ. | Max. | Unit | Test Condition | Note |
|---|---|---|---|---|---|---|---|
| **LED Current** | IF | – | 140 | – | mA | – | – |
| **LED Voltage** | VF | 18.0 | 18.6 | 19.5 | V | – | – |
| **Life Time** | | – | 15000 | – | Hr. | I≦140mA | – |
| **Brightness** | Luminous | 450 | 500 | -- | Nits | | -- |
| **Color** | White | | | | | | |

Note (1) Permanent damage may occur to the LCD module if beyond this specification. Functional operation should be restricted to the conditions described under normal operating conditions.

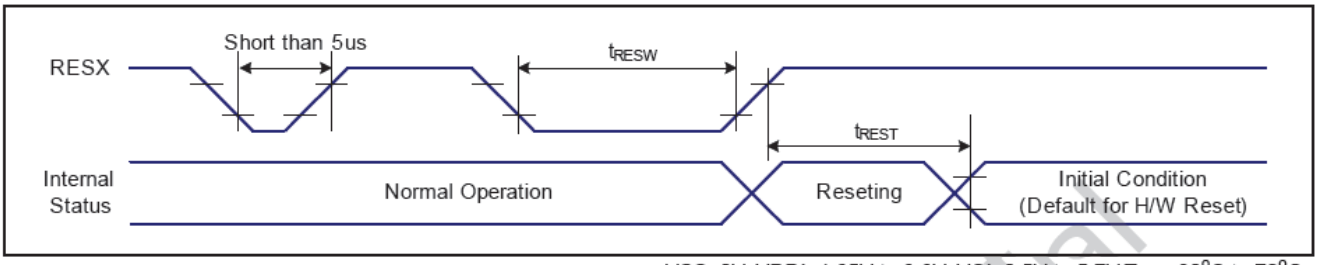
(2)Ta=25±2℃

(3)Test condition: LED Current 140mA



# 5. SEQUENTIAL CHART

## 5.1 RESET TIMING CHARACTERISTICS




VSS=0V, VDDI=1.65V to 3.6V, VCI=2.5V to 5.7V,Ta = -30°C to 70°C

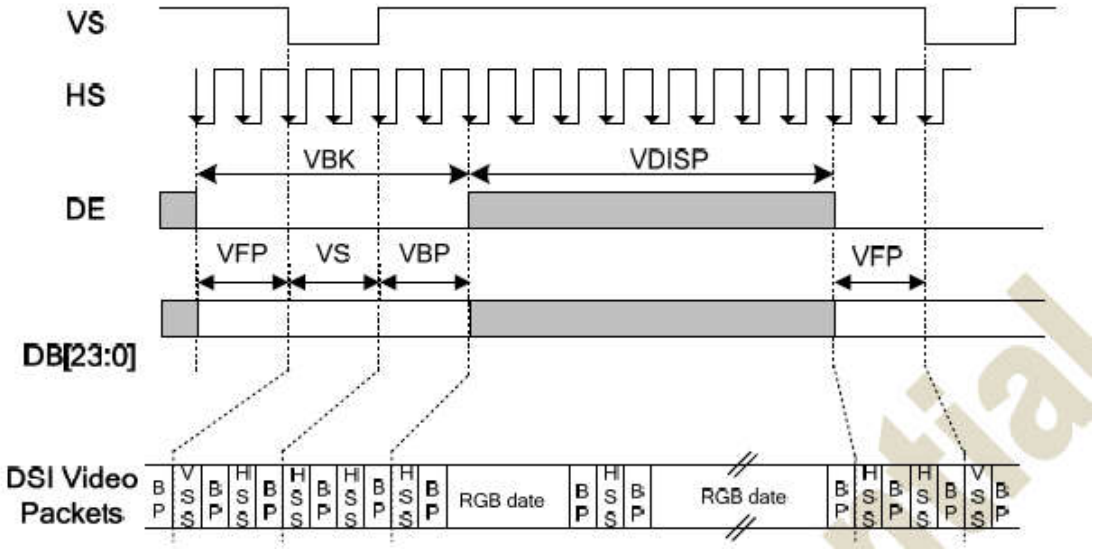
| Symbol | Parameter | Related Pins | MIN | TYP | MAX | Note | Unit |
|---|---|---|---|---|---|---|---|
| $t_{RESW}$ | *1) Reset low pulse width | RESX | 10 | - | - | - | us |
| $t_{REST}$ | *2) Reset complete time | - | - | - | 5 | When reset applied during Sleep in mode | ms |
| | | - | - | - | 120 | When reset applied during Sleep out mode | ms |

Table:Reset input timing

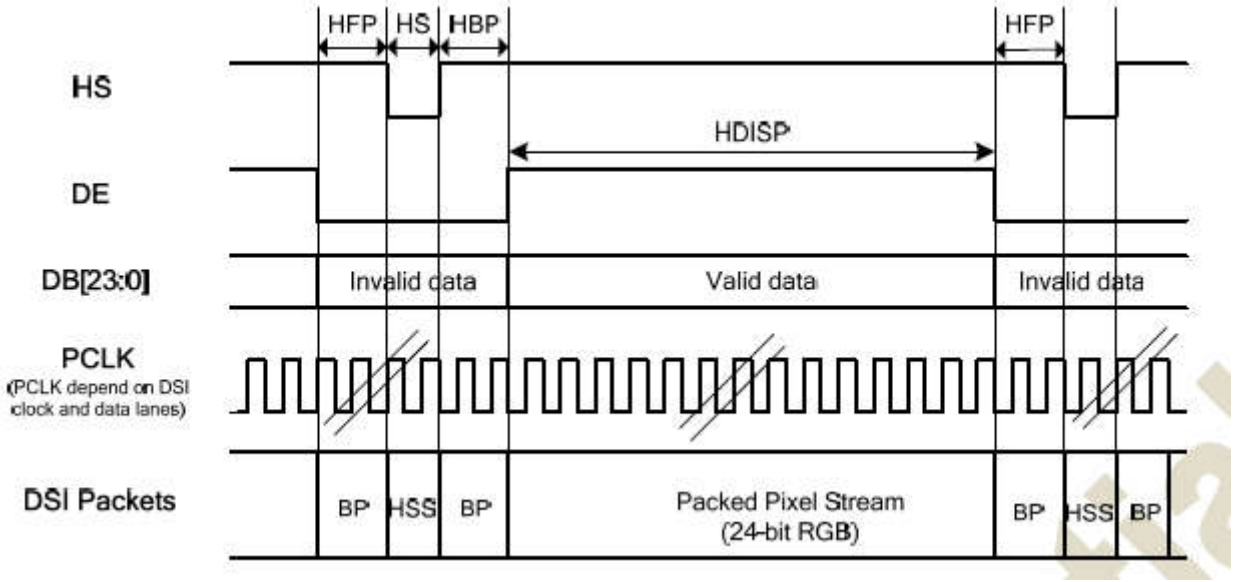
Note 1: Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below.

## 5.2 TIMINGS FOR DSI VIDEO MODE

Vertical Timings



Horizontal Timings

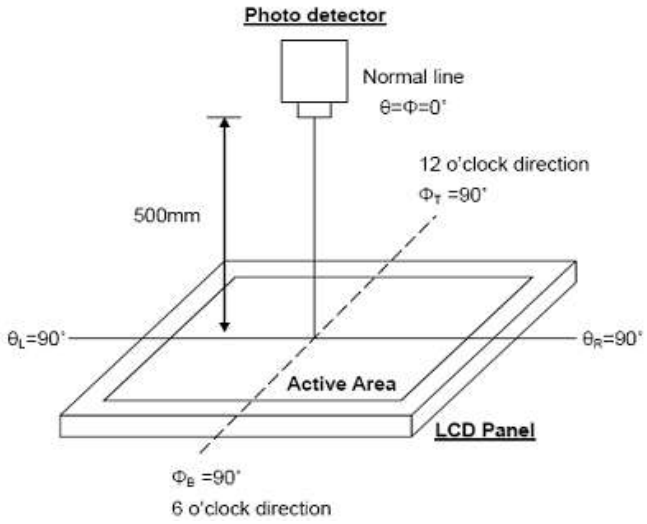


# 6.OPTICAL CHARACTERISTICS

Ta=25℃±2

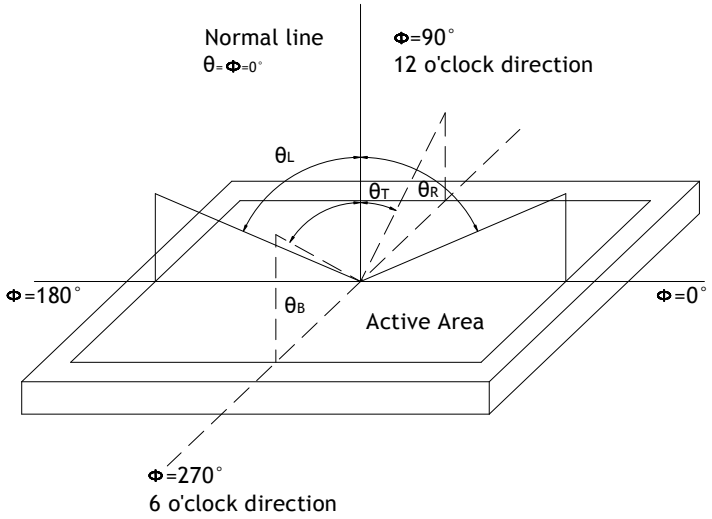
| Item | | Symbol | Condition | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|---|---|
| Contrast Ratio | | CR | Θ = 0° | 600 | 800 | - | | Note1<br>Note4 |
| Luminance | | YL | | | 500 | - | cd/m2 | Note1<br>Note6<br>Note7 |
| Luminance Uniformity | | IV-M | | 70 | 75 | | % | |
| Response Time<br>(Rising + Falling) | | TRT | Ta= 25℃<br>Θ = 0° | - | 30 | 35 | ms | Note1<br>Note3 |
| Viewing<br>Angle range | Horizontal | ΘL | CR > 10 | - | 80 | - | | Note2 |
| | | ΘR | | - | 80 | - | | |
| | Vertical | ΘU | | - | 80 | - | | |
| | | ΘD | | - | 80 | - | | |
| Color<br>Chromaticity | White | x | | -0.03 | 0.319 | +0.03 | | Note1<br>Note5<br>Note7 |
| | | y | | | 0.362 | | | |
| | Red | x | | | 0.652 | | | |
| | | y | | | 0.329 | | | |
| | Green | x | | | 0.340 | | | |
| | | y | | | 0.593 | | | |
| | Blue | x | | | 0.138 | | | |
| | | y | | | 0.109 | | | |
| NTSC | | | | | 58 | | % | |

Note1: Definition of optical measurement system




Note2: Definition of viewing angle range and measurement system

Viewing angle is measured at the center point of the LCD by CONOSCOPE (ergo-80).




Note3: Definition of Response time

The response time is defined as the LCD optical switching time interval between "White" state and "Black" state. Rise time (TON) is the time between photo detector output intensity changed from 90% to 10%. And fall time (TOFF) is the time between photo detector output intensity changed from 10% to 90%.

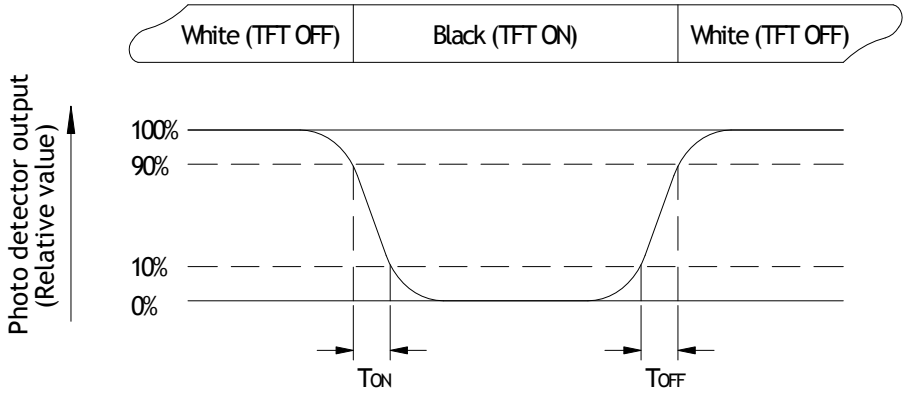


Fig. 6-3 Definition of response time

Note4: Definition of contrast ratio

$$\text{Contrast ratio(CR)} = \frac{\textbf{Luminance measured when LCD on the Whitestate}}{\textbf{Luminance measured when LCD on the Blackstate}}$$

"White state ": The state is that the LCD should drive by Vwhite.

"Black state": The state is that the LCD should drive by Vblack.

Vwhite: To be determined Vblack: To be determined.

Note5: Definition of color chromaticity (CIE1931)

Color coordinates measured at center point of LCD.

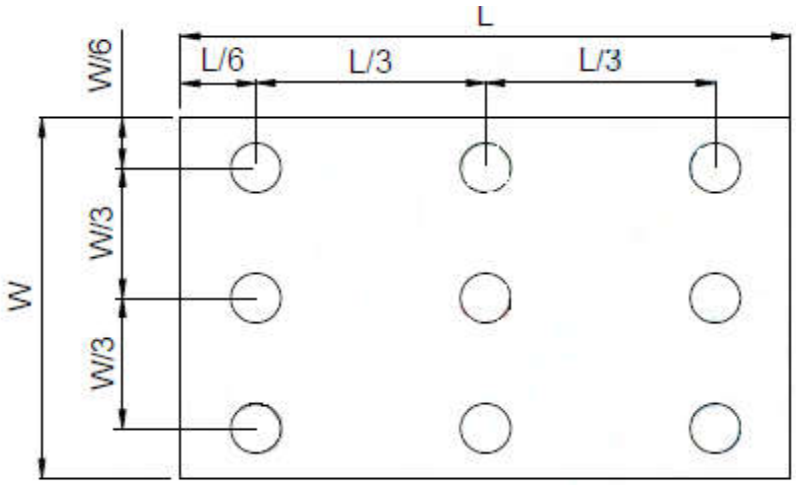
Note6: All input terminals LCD panel must be ground wile measuring the center area of the panel. The LED driving condition is IL=180mA.

Note7: Definition of Luminance Uniformity

Active area is divided into 9 measuring areas.Every measuring point is placed at the center of each measuring area.

Luminance Uniformity (U) = Lmin/ Lmax

L----Active area length, W---- Active area width




Bmax: The measured maximum luminance of all measurement position.

Bmin: The measured minimum luminance of all measurement position.

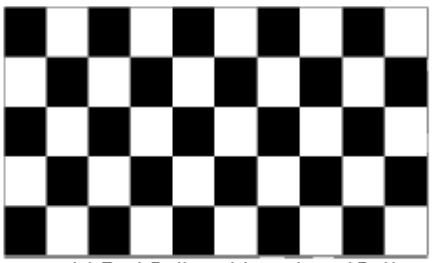
# 7. RELIABILITY TEST ITEMS

## 7.1 TEMPERATURE AND HUMIDITY

| Test Item | Test Condition | Remark |
|---|---|---|
| High Temperature Storage | Ta=60℃; 72hrs | IEC60068-2-1：2007<br>GB2423.2-2008 |
| Low Temperature Storage | Ta=-20℃; 72hrs | IEC60068-2-1：2007<br>GB2423.1-2008 |
| High Temperature Operation | Ta=50℃; 72hrs | IEC60068-2-1：2007<br>GB2423.2-2008 |
| Low Temperature Operation | Ta=0℃; 72hrs | IEC60068-2-1：2007<br>GB2423.1-2008 |
| High Temperature High Humidity Operation | Ta=50℃，90%RH，<br>72Hrs(no condensation) | IEC60068-2-78：2001<br>GB/T2423.3-2006 |
| Thermal Shock | -20℃(0.5h) ~ 60℃(0.5h)<br>/ 10cycles | Start with cold temperature，<br>End with high temperature，<br>IEC60068-2-14:1984,GB2423.22-2002 |
| Image Sticking | 25℃ ; 2hrs | Note1 |

Note1:Condition of image sticking test :25℃±2℃

Operation with test pattern sustained for 2hrs,then change to gray pattern immediately.after5 mins,themura must be disappeared completely



(a) Test Pattern (chess board Pattern )



(b) Gray Pattern

## 7.2ESD

| Test item | Conditions | Remark | |
|---|---|---|---|
| Electro Static Discharge Test (non-operation) | 150pF，330Ω，<br>Contact:±4KV,Air:±8KV | 1 | IEC61000-4-2：2001<br>GB/T17626.2-2006 |
| | 200pF，0Ω，±200V contact test | 2 | |

Note: Measure point :

1. LCD glass and metal bezel
2. IF connector pins

# 8. General Precaution

## 8.1 Safety

1. Do not swallow any liquid crystal, even if there is no proof that liquid crystal ispoisonous.
2. If the LCD panel breaks, be careful not to get liquid crystal to touch your skin.
3. If skin is exposed to liquid crystal, wash the area thoroughly with alcohol or soap.

## 8.2 Storage Conditions

1. Store the panel or module in a dark place where the temperature is 23±5° C andThehumidity is below 50±20%RH.
2. Store in anti-static electricity container.
3. Store in clean environment, free from dust, active gas, and solvent.
4. Do not place the module near organics solvents or corrosive gases.
5. Do not crush, shake, or jolt the module.

## 8.3 Handling Precautions

(1) Avoid static electricity which can damage the CMOS LSI.
(2) The polarizing plate of the display is very fragile. So, please handle it verycarefully.
(3) Do not give external shock.
(4) Do not apply excessive force on the surface.
(5) Do not wipe the polarizing plate with a dry cloth, as it may easily scratch theSurface of plate.
(6) Do not use ketonics solvent & Aromatic solvent, use with a soft cloth soaked witha cleaning naphtha solvent.
(7) Do not operate it above the absolute maximum rating.
(8) Do not remove the panel or frame from the module.
(9)When the module is assembled, it should be attached to the system firmly,Be careful not to twist and bend the module.
(10)Wipe off water droplets or oil immediately. If you leave the droplets for a long time,staining and discoloration may occur.
(11) If the liquid crystal material leaks from the panel,it should be kept away from theeyes or mouthIn case of contact with hands,legs or clothes, it must be washedawaythoroughly with soap.

## 8.4 Warranty

(1)The period is within twelve months since the date of shipping out under normal using and storage conditions.
(2) Do not repaired or modified the LCM. It may cause function to lose efficacy,HENGJIAYU does not warrant the LCM.
(3) All process and material comply ROHS.

# 9. Package Drawing

TBD